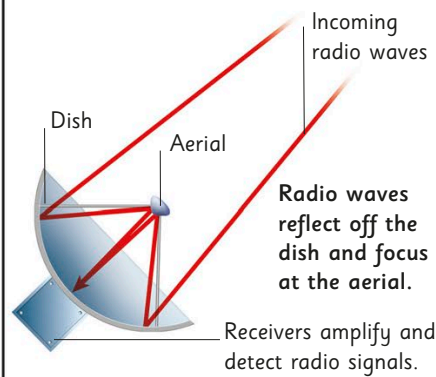


How a radio telescope works

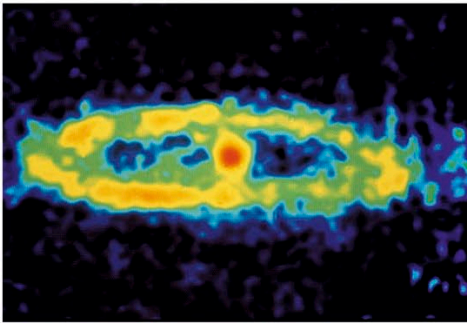


Radio telescopes

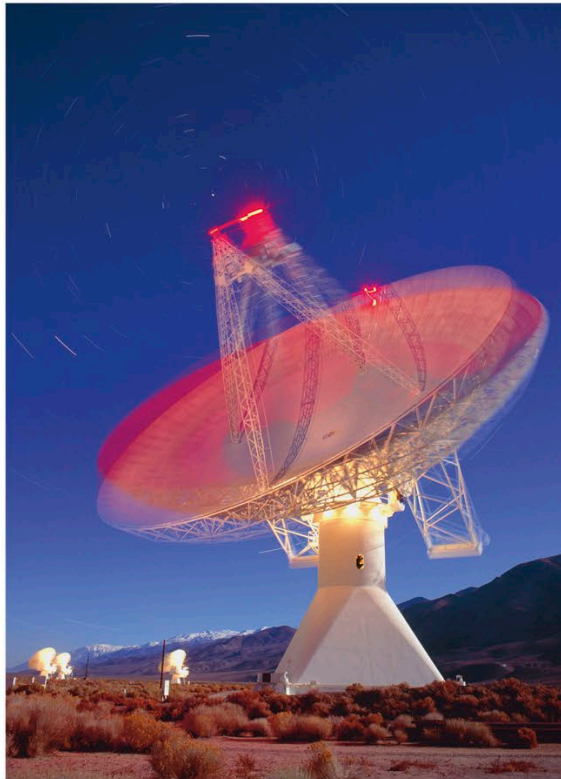
Invisible radio waves surround us. They also reach us from space, and large dishes are used to pick them up, day and night, to help astronomers learn more about space.

So do the dishes “listen” to space?

No. Radio astronomers do not listen to noises. Sound waves do not pass through space. The dishes pick up radio waves, a receiver measures them, and a computer turns this information into a picture.



Radio telescope image of the Andromeda Galaxy. The red centre is producing the strongest signals.



Tilt and learn

Radio dishes are designed to tilt and move around, so radio astronomers can point them at the bit of space they want to study. Also, as the Earth turns, radio dishes need to move in order to follow one spot in the sky. The movements, which are controlled by computer, are very precise.

No mountain in sight!

Unlike optical telescopes, radio telescopes don't need to be built at the top of mountains, as radio waves will pass through cloud cover. Each dish reflects and focuses the incoming radio signals onto an aerial mounted above it. The dishes can be enormous.

